

Insulin Dose Calculator — Parameter Grid (Ranges)

Pediatric rapid-acting insulin · clinical decision support · companion to the calculator

Each parameter shown as a single allowed range. Rounding is always **up** (ceiling) to the listed increment.

ORDER PARAMETERS — ALLOWED RANGES

Parameter	U-100 Standard (Units)	U-10 Dilute (Syringe Units)
Insulin : carb ratio	1–100 g/Unit	1.5–40 g/Syr Unit
Correction factor (ISF)	4–300 mg/dL/Unit (verify if > 150)	10–200 mg/dL/Syr Unit
Target blood glucose	70–200 mg/dL	120–300 mg/dL
Rounding increment	0.5 or 1 Unit (prescriber-set)	1 Syr Unit (fixed)

ORDER-SIGN SOFT ADVISORIES — BY AGE

U-100 Standard only — U-10 Dilute has no order-sign soft advisories.

Age band	U-100 Standard (Units)		
	I:C low (g/Unit)	I:C high (g/Unit)	CF low (mg/dL/Unit)
0–5 y	below 10	100 or more (all ages)	below 50
5–11 y	below 4		below 15
12 y and older	below 2		below 8

CALCULATOR PARAMETERS — ENTRY GUARDS

Parameter / input	U-100	U-10
Target BG · < 5 y	130–200 mg/dL	—
Target BG · 5–11 y	70–180 mg/dL	—
Target BG · 12 y and older	70–150 mg/dL	—
Carbohydrates eaten	0–300 g; Alerted Calculation cap of 250 g	0–99 g
Current blood glucose	1–2000 mg/dL; alerted cap at 600	1–2000 mg/dL; alerted cap at 600

BLOOD-GLUCOSE DECISION THRESHOLDS (both concentrations)

Condition	Behavior
BG = 0	Critical hypoglycemia — functionally never triggered (entry minimum is 1)
BG < 70	Hypoglycemia — all dosing suppressed
BG > target	Correction applies (otherwise prandial only)
BG > 600	Correction calculated at 600 cap; prandial unaffected

- U-100 target BG is trapped at the harmonized 70–200 mg/dL at order entry; the finer age-tiered limits are enforced at the dose calculator by age check (rows above).
- U-10 target BG (120–300 mg/dL, single infant band) is the same at order entry and calculator.
- Correction-factor “verify if > 150” is a soft advisory for U-100 only; U-10 has no advisory.
- Order-sign soft low advisories (U-100, by age) are non-blocking — they fire at signing when a hard-valid value sits below the age-typical floor; the prescriber acknowledges to proceed. The I:C high advisory fires at 100 (the maximum), flagging sliding-scale-style orders where a very high ratio makes prandial coverage negligible; a CF high warn at 200 is planned.
- Current BG entry is 1–2000 mg/dL; correction uses up to 600 (alerted). BG = 0 cannot be entered, so critical hypoglycemia is functionally never triggered at the calculator.
- Out-of-range parameters are corrected to the nearest boundary; mL volume is never displayed.
- Reference only — not for clinical use without pharmacy & informatics committee review.